1 Notation and conventions

Indices and sets. We use lower-case Latin letters for indices and calligraphic upper-case for sets:

- $r \in \mathcal{R}$ — receiver. Single-receiver modes (PPP, PPP-RTK) have $|\mathcal{R}| = 1$; baseline modes have a fixed reference and one or more rovers; the network producer (`PPP_RTK_Network`) carries a peer set of $|\mathcal{R}| \geq 2$ stations (150–305 in practice) with no reference/rover asymmetry.
- $g \in \mathcal{G} = \{\text{GPS}, \text{GAL}, \text{BDS}, \text{GLO}, \text{QZS}, \dots\}$ — constellation.
- $i \in \mathcal{S}_g$ — satellite of constellation g. p_g denotes the pivot satellite of constellation g under the BSSD parameterisation (Section 4).
- $s \in \mathcal{B}$ — signal/observation type, identified by its RINEX-3 three-character code (e.g. `C1C`, `L2W`, `L5X`).
- $k \in \mathbb{N}$ — discrete epoch index.

Vectors and matrices. Bold lower-case for vectors ($\boldsymbol{x}$, $\boldsymbol{r}$); bold upper-case for matrices ($\boldsymbol{H}$, $\boldsymbol{P}$). $\boldsymbol{a}^{\mathrm{T}}$ is the transpose. A hat denotes a state estimate; a tilde a one-step prediction: $\hat{\boldsymbol{x}}_{k|k}$ vs. $\tilde{\boldsymbol{x}}_{k|k-1}$.

Observables. For each receiver/satellite/signal triple, the engine consumes two real-valued observables (in metres) per epoch:

- $P^i_{r,s}$ — pseudorange (code), in metres.
- $\Phi^i_{r,s}$ — carrier phase, scaled to metres via λ_s. Beyond the range, clock, and atmosphere terms it shares with the code, its ambiguous (non-range) part is $\lambda_s N^i_{r,s}$ plus the fractional phase biases $b_{r,s} - b^i_s$ (Equation (5)).

Where the distinction matters, s may further denote the band ($L_1, L_2, L_5, \dots$) and the tracking attribute (e.g. `C`, `W`, `X`). The leading observable-type letter (`C`/`L`) selects code vs. phase; the frequency-only quantities f_s, λ_s, and $N^i_{r,s}$ depend only on the (band, attribute) part, so a code and its sibling phase share them. This matches the state keys, which carry the full phase code for ambiguities (`"G01_L1C"`) but only the band digit for receiver code biases (`"GPS_IFB2"`).

Frequencies and wavelengths. f_s is the carrier frequency (Hz) and $\lambda_s = c/f_s$ the wavelength (m). The frequency-squared scaling factor used to project the L1-equivalent slant ionospheric delay onto signal s is

$$\mu_s \;:=\; \frac{f_1^2}{f_s^2}. \tag{1}$$

Here f_1 is the GPS L1 frequency (1575.42 MHz), used as the common reference for *every* constellation — not the band-1 frequency of the satellite's own system. The estimated state $I^i_{r,1}$ is therefore GPS-L1-equivalent even for BDS/GLO/GAL satellites (`MeasurementModel::buildJacobianRow`: `f_ref = Constants::GPS_L1_FREQ`).

Differencing operators. Three differencing operations recur throughout the document:

$$\Delta q^{i,p_g}_{r,s} := q^i_{r,s} - q^{p_g}_{r,s}, \quad \nabla q^i_{r_1 r_2,s} := q^i_{r_1,s} - q^i_{r_2,s}, \quad \nabla\Delta q^{i,p_g}_{r_1 r_2,s} := \Delta\big(\nabla q^i_{r_1 r_2,s}\big).$$

Read Δq as "between-satellite single difference of q relative to pivot p_g," ∇q as "between-receiver single difference," and $\nabla\Delta q$ as "double difference."

Sign conventions. We adopt the convention that the slant ionospheric delay $I^i_{r,s}$ enters *additively* on the code observable and *subtractively* on the carrier phase observable; this is the source of the well-known "code–phase divergence." All other range-additive terms (geometry, clocks, troposphere, biases) carry the same sign on both observables. See Equations (4) and (5) for the explicit equations.

Symbol table.

Symbol	Units	Meaning
$\boldsymbol{r}_r, \boldsymbol{r}^i$	m	ECEF position of receiver/satellite
ρ^i_r	m	geometric range $\|\boldsymbol{r}^i - \boldsymbol{r}_r\|$
$\boldsymbol{e}^i_r$	—	unit LOS from r to i in ECEF
$c\,\mathrm{d}t_r$	m	receiver clock offset (scaled by c)
$c\,\mathrm{d}t^i$	m	satellite clock offset (CLK product in user modes; estimated state `"<sat>|SC"` in `PPP_RTK_Network`)
$c\,\delta t^{(g)}_r$	m	receiver inter-system time offset for constellation $g \neq g_{\mathrm{ref}}$
T^i_r	m	total tropospheric path delay
$\mathrm{ZHD}^i_r, \mathrm{ZWD}^i_r$	m	zenith hydrostatic / wet components
$I^i_{r,1}$	m	L1-equivalent slant ionospheric delay
$N^i_{r,s}$	cyc	integer carrier-phase ambiguity
$b_{r,s}, b^i_s$	m	receiver / satellite phase biases
$\bar{b}^i_s$	m	estimable satellite phase bias (network mode; Section 2)
$B_{r,s}, B^i_s$	m	receiver / satellite code biases (OSB)
$B^{(g,b)}_{r,k}$	m	per-FCN-slot GLONASS receiver IFB, band b; here $k \in [-7, +6]$ is the signed FDMA channel number (a context-local reuse, not the epoch index)
$\varepsilon^P, \varepsilon^\Phi$	m	residual (multipath + thermal)
$\boldsymbol{v}_r$	m/s	receiver velocity (optional / IMU state)
$\lambda_{\mathrm{WL}},\ \lambda_{\mathrm{NL}}$	m	wide- / narrow-lane wavelength $c/(f_1 \mp f_2)$
$\boldsymbol{C}^e_b$	—	body-to-ECEF attitude (direction-cosine) matrix

2 State vector layout

The Kalman filter operates on a state vector $\boldsymbol{x}_k \in \mathbb{R}^{n_k}$ whose dimension n_k *varies across epochs*: per-satellite and per-(satellite, signal) substates are added when a new sat/signal is first observed and pruned (or marked stale) when the satellite drops below the elevation mask or its phase track is interrupted by a cycle slip without recovery. All bookkeeping — key→index maps, free-list re-use, and dimensional reshaping of $\boldsymbol{x}$ and $\boldsymbol{P}$ — is encapsulated in `StateManager`; `KalmanFilter` consumes the state through that interface and never indexes raw memory by hard-coded offsets.

2.1 Schematic layout

In schematic form, suppressing the epoch index k:

$$\begin{aligned}\boldsymbol{x} \;=\; \big[\; & \underbrace{\boldsymbol{r}_r^{\mathrm{T}}}_{3},\; \underbrace{\boldsymbol{v}_r^{\mathrm{T}}}_{0\text{ or }3},\; \underbrace{c\,\mathrm{d}t_r}_{1},\; \underbrace{(c\,\delta t_r^{(g)})_{g\neq g_{\mathrm{ref}}}^{\mathrm{T}}}_{|\mathcal{G}|-1},\\ & \underbrace{\mathrm{ZWD}_r,\,(g_N,g_E)}_{1+0\text{ or }1+2},\; \underbrace{(c\,\mathrm{d}t_r^{\phi})}_{0\text{ or }1},\\ & \underbrace{(I_{r,1}^i)_{i\in\mathcal{V}_k}^{\mathrm{T}}}_{|\mathcal{V}_k|},\; \underbrace{(N_{r,s}^i)_{(i,s)\in\mathcal{A}_k}^{\mathrm{T}}}_{|\mathcal{A}_k|},\\ & \underbrace{(B_{r,s})_{s\in\mathcal{B}}^{\mathrm{T}}}_{|\mathcal{B}|},\; \underbrace{(B_s^i)_{(i,s)\in\mathcal{D}_k}^{\mathrm{T}}}_{|\mathcal{D}_k|}\;\big]^{\mathrm{T}}. \end{aligned} \tag{2}$$

where

- $\mathcal{V}_k$ is the set of satellites currently visible at epoch k (i.e. above the elevation mask, with valid orbit/clock products and at least one tracked signal),
- $\mathcal{A}_k \subseteq \mathcal{V}_k \times \mathcal{B}$ is the set of phase-tracked (satellite, signal) pairs,
- $\mathcal{D}_k \subseteq \mathcal{V}_k \times \mathcal{B}$ is the subset of (satellite, signal) pairs for which the BIA product lacks a satellite OSB and a state-space estimate is therefore created (see `estimation.estimate_sat_biases`).

For brevity the schematic suppresses several conditional refinements detailed below: the rate states that accompany velocity (a receiver clock-rate and one inter-system clock-rate offset per non-reference constellation), the optional per-satellite slant-iono *rate* companion under `enable_iono_rate`, the network-producer satellite-clock and receiver-phase-bias states, the nine-state inertial augmentation of the inertial-aiding document, and the fact that the entire receiver-side clock and inter-system-offset block is removed under the BSSD parameterisation (Section 4.1).

2.2 Component-by-component description

Position $\boldsymbol{r}_r \in \mathbb{R}^3$. ECEF Cartesian receiver position. Always present. Initialised either from the RINEX header approximate position or, after the first epoch, from a Weighted Least-Squares (WLS) seed solution to within the linearisation radius of the geometric range model (Section 3.3).

Velocity $\boldsymbol{v}_r \in \mathbb{R}^3$. Optional. Enabled by `estimation.estimate_velocity = true`, or forced on by tightly-coupled IMU integration. When present, position and velocity form a coupled continuous-white-noise-acceleration (CWNA, integrated-random-walk) block driven by a *single* acceleration spectral density `process_noise.acceleration_psd_q` (the CWNA process-noise model in the estimation document); the scalar `position_q` of the velocity-free case is then unused. Enabling velocity also introduces a receiver clock-rate state $c\,\dot{\mathrm{d}}t_r$, and one inter-system clock-rate offset per non-reference constellation, each CWNA-coupled to its bias counterpart (`clock_rate_q`, `system_time_rate_offset_q`).

Receiver clock $c\,\mathrm{d}t_r$. Single state, in metres. Process noise is large (`clock_bias_q` $\sim 10^4\,\mathrm{m}^2$ per second) so the state is effectively re-estimated each epoch; this captures oscillator drift without imposing a stochastic model that may not match the receiver's actual oscillator. Under the BSSD (between-satellite single-difference) parameterisation (Section 4.1) the receiver clock is unobservable

and the state is dropped entirely — as are the inter-system offsets below — so the whole receiver-side clock block vanishes in PPP-RTK SD mode.

Inter-system time offsets $c\,\delta t_r^{(g)}$. One per active constellation other than the chosen reference (default GPS). These absorb the per-constellation clock biases that arise from mixing analysis-centre-defined clock products across systems ("GAL − GPS time offset," etc.).

Tropospheric ZWD and gradients. ZWD_r models the zenith wet delay; the zenith hydrostatic component ZHD_r is treated deterministically via the Saastamoinen model [3]. Total slant tropospheric delay is then

$$T_r^i \;=\; m_h(\theta_r^i)\,\mathrm{ZHD}_r \;+\; m_w(\theta_r^i)\,\mathrm{ZWD}_r \;+\; m_w(\theta_r^i)\cot\theta_r^i\big[g_N\cos\alpha_r^i + g_E\sin\alpha_r^i\big], \tag{3}$$

with elevation θ_r^i, azimuth α_r^i, and the Niell or VMF mapping functions m_h, m_w. Gradients (g_N, g_E) are optional.

Per-satellite slant ionosphere $I_{r,1}^i$. PPP-RTK-only. One state per visible satellite, expressed at the L1 frequency; signal-specific delays follow Equation (1). Initial value seeded from the Klobuchar broadcast model when no external ionospheric product is available. Its process noise is a random walk whose zenith spectral density `ionosphere_q` is projected onto the slant by the squared obliquity factor $1/\sin^2\theta_r^i$ (the ionospheric process-noise model in the estimation document), letting low-elevation states vary faster. Modelling these per-sat states explicitly is what permits PPP-RTK to retain integer ambiguity recovery without forming the ionosphere-free combination (which amplifies code noise by a factor of ~ 3). Optionally, under `estimation.enable_iono_rate` (default off), each visible satellite carries a companion constant-velocity block $[\,I_{r,1}^i,\ \dot{I}_{r,1}^i\,]$: the slant-iono *rate* $\dot{I}_{r,1}^i$ is unobserved directly ($H = 0$) and driven only by the F-coupling into the level state, with its own spectral density `process_noise.iono_rate_psd` and its own slip response (`reinflateIonosphereRateVariance`), letting the filter track a linear slant ramp rather than a pure random walk.

Carrier-phase ambiguities $N_{r,s}^i$. One state per phase-tracked (i, s) pair, in metres (the state value is λN; the `nl_wl` basis below is the exception, carrying integers in cycles). Treated as float during the prediction/update cycle and resolved to integers post-hoc by the LAMBDA-family algorithms in `filter/AmbiguityResolver` [5] when AR is enabled. Cycle slips trigger a re-seed via `StateManager::resetAmbiguity`, which *decorrelates* the state rather than repairing it: the ambiguity's covariance row and column are zeroed and its diagonal reset to σ_m^2, while the state *value* is left unchanged (it is overwritten by the geometry-based seed in `ensureRequiredStates`). This follows the project's reset-not-repair convention — a slipped ambiguity carries no trustworthy correlation to the rest of the state, so the cross-state correlations are broken, not preserved.

Receiver code biases $B_{r,s}$. These absorb the receiver-side inter-frequency biases that the OSB convention does not reference to zero. They are *not* one per signal: `StateInitializer::resolveReceiverBias` keys them per (system, frequency band) as inter-frequency biases (`"<system>_IFB<band>"`) plus, in the alternative scheme, a single per-system differential code bias (`"<system>_DCB"`). Like the receiver clock they are receiver-common and so are suppressed under BSSD.

Per-slot GLONASS inter-frequency bias $B_{r,k}^{(g,b)}$. Optional, off by default. GLONASS FDMA assigns each slot $k \in [-7, +6]$ a distinct carrier frequency $f = f_{\text{base}} + k\,\Delta f$, so the receiver group delay differs *per slot* and a single per-band IFB state cannot represent it. When `estimation.glonass_ifb.enable = true` (and the prerequisite `estimation.estimate_ifb`; the loader warns and the flag is inert otherwise), `resolveReceiverBias` instead emits one state per (FDMA band $b \in \{\text{G1}, \text{G2}\}$, slot k) with unit measurement coefficient, keyed `"GLONASS_IFB<band>_k<±slot>"`. The central channel $k = 0$ is the held datum and produces no state; its constant is absorbed by the GLONASS inter-system time offset. CDMA bands (G3/G4/G6) have a slot-independent frequency and retain the ordinary per-band treatment above. Crucially, unlike the receiver-common biases these columns *survive* between-satellite single differencing — two GLONASS satellites on different slots do not cancel — so `BssdTransform` and `StateManager` special-case them (via `isGlonassPerSlotIfbKey`) to exempt them from the receiver-block zeroing; only the inter-slot *differences* remain estimable in SD.

Satellite code biases B_s^i. Per (satellite, signal) pair, but *only when the BIA file does not supply an OSB* for that pair — typically a third-frequency signal in a triple-frequency setup. When present, these states are weakly stochastic (`sat_bias_q` $\sim 10^{-10}$) since the underlying physical biases are nearly stationary on the daily timescale.

Satellite phase biases $\bar{b}_s^i$ (network mode). Present only in the network producer (`ProcessingMode::PPP_RTK_Network`) and gated by `estimation.estimate_sat_phase_biases` (default off, inert in every other mode). The user side imports the satellite phase bias as a known SSR correction (the SSR-corrections document) or absorbs it into the float ambiguity; the network instead carries an *explicit* per-(satellite, signal) phase-bias state on the undifferenced phase rows (the estimable bias of Odijk [2], lumping the network pivot-receiver ambiguity). On a single station's row it is co-linear with the per-receiver float ambiguity and is separable only *across* stations in the joint network estimator (the network-mode document). It lives in a *distinct* state map from the satellite *code* biases above (`StateManager::sat_phase_bias_indices_`), keyed `"<sat>|<signal>|PB"`, and carries near-stationary process noise (`sat_phase_bias_q` $\sim 10^{-12}$).

Receiver phase biases $\delta_{r,s}$ (network mode). The receiver-side counterpart of the satellite phase bias, present only in the network producer and gated by `network.phase_datum_eq34` (the Eq-34 / single-receiver-chain phase datum; see the network-mode chapter). Giving the receiver fractional phase bias its *own* estimable state $\delta_{r,s}$ stops it lumping into the emitted satellite phase bias, so the emitted between-satellite reference stays integer-consistent across *all* satellites (each chain hop then contributes only an integer). It is per (receiver, system, signal), keyed `"<system>_RPB_<signal>"` in its own map (`StateManager::receiver_phase_bias_indices_`). Structurally it is unlike every other bias block described here: these are *local* states (each observed by a single station) that are *re-estimated fresh every epoch* — no process noise, no cross-epoch carry. The receiver-side wander sinks into $\delta_{r,s}$ while the time-linked integer information stays in the ambiguity carry.

Satellite clocks $c\,\Delta\delta t^i$ (network mode). Also network-producer-only, gated by `estimation.estimate_sat_clocks` (default off, inert in every other mode). The network estimates an *explicit* per-satellite clock correction $\Delta\delta t^i$ (in metres, relative to the a-priori precise/real-time product clock already removed from `obs.modeled`) so the emitted clocks stay consistent with the chosen S-system datum. It is *per satellite*, not per signal — one state per satellite, with no signal field

— and the satellite *orbit* stays fixed from the product; only the clock is estimated. It lives in a distinct map (`StateManager::sat_clock_indices_`), keyed `"<sat>|SC"`, with near-white process noise (`sat_clock_q` $\sim 10^{-2}$).

Inertial augmentation. When a tightly-coupled IMU is active the core state grows by nine further entries — a three-axis attitude error and the accelerometer and gyroscope biases — propagated by the strapdown mechanisation rather than the random-walk process model. These are treated in full in the inertial-aiding document.

Baseline differential troposphere. In baseline mode the per-station tropospheric, clock, ionosphere, and bias states are absent; an optional single differential zenith-wet-delay state $\Delta\mathrm{ZWD}_{r_{\mathrm{rov}}r_{\mathrm{ref}}}$ may be carried (`baseline.estimate_differential_tropo`, Section 4.3).

2.3 Dynamic indexing

The variable-dimension blocks are addressed via key→index maps maintained by `StateManager`. The canonical key formats are defined once in `include/filter/StateKeySchema.hpp`, each with a `make`/`parse` pair:

- iono key: `"<sys><prn>"`, e.g. `"G05"` (delimiter-free);
- ambiguity key: `"<sys><prn>_<signal>"`, e.g. `"G05_L1C"` — split on the *first* underscore, so an iono-free combination `"G05_L1C_L5Q"` retains its compound signal;
- satellite signal-bias key (code): `"<sys><prn>|<signal>"`, e.g. `"G05|C1W"` — a *pipe* delimiter and a distinct map, *not* the underscore ambiguity namespace;
- satellite phase-bias key (network): `"<sys><prn>|<signal>|PB"`, e.g. `"G05|L1C|PB"` — the trailing `|PB` keeps it disjoint from the code-bias namespace, in its own map (`sat_phase_bias_indices_`);
- receiver phase-bias key (network): `"<system>_RPB_<signal>"`, e.g. `"Galileo_RPB_L5Q"` — the `_RPB_` infix keeps it disjoint from the receiver code-bias namespace (which matches on `IFB`/`DCB`), in its own map (`receiver_phase_bias_indices_`);
- satellite clock key (network): `"<sat>|SC"`, e.g. `"G05|SC"` — per satellite, *not* per signal; the trailing `|SC` discriminator gives the clock its own map (`sat_clock_indices_`);
- receiver code-bias key: `"<system>_IFB<band>"` or `"<system>_DCB"`; the per-slot GLONASS variant appends an FCN marker, `"GLONASS_IFB<band>_k<±slot>"` (e.g. `"GLONASS_IFB1_k-4"`);
- inter-system time-offset key: the bare system name (`"Galileo"`, …; GPS is the reference and never appears);
- double-difference ambiguity key (baseline): `"DD|<cons>|<sat>|<code>|<pivot>"`.

This single source of truth eliminates the off-by-one errors that plague hand-managed dynamic state vectors and makes the BSSD/DD S-transforms (Section 4) implementable as block operations on the relevant key sets.

Ambiguity basis. By default the ambiguity block carries one per-band float ambiguity per tracked (satellite, signal) in metres. Setting `estimation.state_ambiguity_basis = "nl_wl"` switches the whole run to a narrow-lane / wide-lane parameterisation: per satellite a narrow-lane state (key `"<sat>_NL_<band>"`) and one wide-lane state per band pair (`"<sat>_WL_<a>_<b>"`, semantically

$N_a - N_b$), both in *cycles* rather than metres so the wide-lane and LAMBDA arithmetic stays native to integers, with the wavelength scaling moved into the Jacobian. The two bases never coexist in a single state vector.

3 Observation equations

This section gives the full *undifferenced* observation model that the filter linearises around at each epoch. The PPP-RTK parameterisation is uncombined: per-signal pseudorange and carrier phase are consumed directly, without forming ionosphere-free or wide-lane combinations as a preprocessing step. Combinations (IF, MW, GF) are still computed for diagnostic and slip-detection purposes (the cycle-slip document; see `slip/MwSlipDetector`, `slip/GraphicSlipDetector`) but they do not enter the measurement update path. This holds whenever the uncombined parameterisation is active; the loader forces `uncombined_enabled=true` for the whole PPP-RTK family (user, network producer, network consumer) in `GnssConfig::loadFromJson`. Plain PPP mode leaves the flag at its default `false` and instead builds ionosphere-free combined observables (keys such as `C1W_C5X`) that *do* feed `MeasurementModel::buildJacobianRow` and the update; the sequel is written for the uncombined case.

3.1 Code (pseudorange) observation

For receiver r tracking satellite i on signal s, the observed pseudorange is

$$\boxed{P^i_{r,s} \;=\; \rho^i_r \;+\; c(\mathrm{d}t_r - \mathrm{d}t^i) \;+\; c\,\delta t_r^{(g(i))} \;+\; T^i_r \;+\; \mu_s\, I^i_{r,1} \;+\; B_{r,s} - B^i_s \;+\; \varepsilon^P_{r,i,s}.} \tag{4}$$

3.2 Carrier-phase observation

The corresponding carrier-phase observation, scaled to metres by λ_s, is

$$\boxed{\Phi^i_{r,s} \;=\; \rho^i_r \;+\; c(\mathrm{d}t_r - \mathrm{d}t^i) \;+\; c\,\delta t_r^{(g(i))} \;+\; T^i_r \;-\; \mu_s\, I^i_{r,1} \;+\; \lambda_s\, N^i_{r,s} \;+\; b_{r,s} - b^i_s \;+\; \varepsilon^\Phi_{r,i,s}.} \tag{5}$$

The defining structural difference between Equations (4) and (5) is the sign of the ionospheric term: $+\mu_s I$ on code, $-\mu_s I$ on phase. Three additional remarks:

1. The phase ambiguity $N^i_{r,s}$ is integer in principle but is contaminated by the receiver-plus-satellite phase biases $b_{r,s} - b^i_s$. In a *float* solution these biases are absorbed into $N^i_{r,s}$, leaving a non-integer state with no structural advantage over the code observation. Recovering integer-valued ambiguities therefore requires either external Fractional Cycle Bias (FCB) products or differencing (Section 4), which cancels common-mode bias terms.
2. The clock product $c\,\mathrm{d}t^i$ is read from the analysis-centre CLK file (post-processing) or refined from the broadcast clock by a streamed SSR correction (real-time, the SSR-corrections document) and treated as deterministic in the filter; any residual product error appears on the observation and is absorbed by the receiver clock state.
3. The system time offset $c\,\delta t_r^{(g(i))}$ is zero by construction when $g(i)$ is the reference constellation (default GPS). Together with the receiver clock $c\,\mathrm{d}t_r$ this means a multi-constellation run carries $|\mathcal{G}|$ clock-like states in total.

3.3 Geometric range and linearisation

The geometric range $\rho_r^i = \|\boldsymbol{r}^i - \boldsymbol{r}_r\|_2$ enters Equations (4) and (5) non-linearly through the receiver position $\boldsymbol{r}_r$. The filter linearises about the current state estimate $\hat{\boldsymbol{r}}_r$:

$$\rho_r^i \;=\; \hat{\rho}_r^i \;-\; (\boldsymbol{e}_{\hat{r}}^i)^{\mathrm{T}}(\boldsymbol{r}_r - \hat{\boldsymbol{r}}_r) \;+\; \mathcal{O}\big(\|\boldsymbol{r}_r - \hat{\boldsymbol{r}}_r\|^2\big)\,, \tag{6}$$

with the unit line-of-sight

$$\boldsymbol{e}_{\hat{r}}^i \;:=\; \frac{\boldsymbol{r}^i - \hat{\boldsymbol{r}}_r}{\|\boldsymbol{r}^i - \hat{\boldsymbol{r}}_r\|_2}. \tag{7}$$

The Jacobian row $\boldsymbol{H}_{(i,s)}$ contributing to the design matrix at the position columns is therefore $-(\boldsymbol{e}_{\hat{r}}^i)^{\mathrm{T}}$. For sub-metre residual targets this single linearisation per epoch is sufficient; the engine optionally relinearises within the epoch (`processing.relinearize_geometry`, which defaults to off in code and is enabled in the PPP, PPP-RTK, baseline, and real-time configs but left off in the six network-producer configs) to handle long initial position offsets.

The satellite position $\boldsymbol{r}^i$ is itself a function of the transmission time. Rather than iterate the emission time against the geometric range — which depends on $\boldsymbol{r}_r$ — `Geometry::computeEmissionState` solves it in a single step from the *pseudorange*: the time of flight is $\tau = P/c$, subtracted (with the satellite clock) from the receiver epoch. Because the receiver clock bias $\mathrm{d}t_r$ enters the receiver epoch as $+\mathrm{d}t_r$ and the pseudorange as $+c\,\mathrm{d}t_r$, it cancels exactly, leaving a residual $(I + T + \varepsilon)/c \approx 30$–$60$ ns of satellite displacement — sub-millimetre, below any PPP threshold. This deliberately avoids a range-based fixed point: iterating against ρ_r^i would re-introduce an uncancelled $\mathrm{d}t_r$ and drive a positive-feedback loop through the filter's receiver-clock state. The only second pass (the `model_corrections.refine_emission_time` block in `Geometry::computeEmissionState`) re-runs the emission solve after subtracting slant tropo and IF-combined ATX from the seed pseudorange — a pseudorange-cleanliness refinement, not a range-versus-position iteration.

3.4 Stochastic model

Observation noise covariance is structured per epoch as a block diagonal matrix with entries

$$\sigma_{r,i,s}^2 \;=\; \sigma_{0,\mathrm{type}(s)}^2 \;\cdot\; w(\theta_r^i)^{-1} \;\cdot\; \xi(\mathrm{C/N_0 \text{ at } } (r,i,s)), \tag{8}$$

combining a per-type baseline ($\sigma_{0,\mathrm{code}}$, $\sigma_{0,\mathrm{phase}}$ from `filter.measurement_noise`), an elevation-dependent down-weighting factor $w(\theta) \in \{1, \sin\theta, \sin^2\theta, \dots\}$ selected by `filter.stochastic_model`, and an optional carrier-to-noise ratio scaling ξ parameterised by `snr_model_params`. Cross-correlations between observations are assumed zero in the undifferenced parameterisation; differencing introduces structured correlations handled in Section 4.

4 Differencing transformations

PPP-RTK and baseline modes do not consume the undifferenced observations of Equations (4) and (5) directly. Each forms a linear combination across satellites (BSSD), receivers (BRSD), or both (DD), absorbing common-mode terms into the cancellation and recovering integer-friendly residual structure. This section gives both the algebraic cancellations and the matrix-form S-transforms that map undifferenced parameters to differenced ones.

4.1 Between-satellite single difference (BSSD)

For each constellation g, choose a pivot satellite p_g (elected by `PivotManager::selectPivotCandidates` under the "sticky pivot" retention policy; the selection ordering is detailed in the *Pivot selection* note below). For any non-pivot satellite $i \in \mathcal{S}_g \setminus \{p_g\}$ and signal s, define

$$\begin{aligned} \Delta P_{r,s}^{i,p_g} &:= P_{r,s}^{i} - P_{r,s}^{p_g}, \\ \Delta \Phi_{r,s}^{i,p_g} &:= \Phi_{r,s}^{i} - \Phi_{r,s}^{p_g}. \end{aligned} \tag{9}$$

Substituting Equations (4) and (5) and cancelling the receiver-side common terms ($c\,\mathrm{d}t_r$, $c\,\delta t_r^{(g)}$, $B_{r,s}$, $b_{r,s}$) yields

$$\Delta P_{r,s}^{i,p_g} = \Delta\rho_r^{i,p_g} - c\,\Delta\mathrm{d}t^{i,p_g} + \Delta T_r^{i,p_g} + \mu_s\,\Delta I_{r,1}^{i,p_g} - \Delta B_s^{i,p_g} + \Delta\varepsilon^{P}, \tag{10}$$

$$\Delta \Phi_{r,s}^{i,p_g} = \Delta\rho_r^{i,p_g} - c\,\Delta\mathrm{d}t^{i,p_g} + \Delta T_r^{i,p_g} - \mu_s\,\Delta I_{r,1}^{i,p_g} + \lambda_s\,\Delta N_{r,s}^{i,p_g} - \Delta b_s^{i,p_g} + \Delta\varepsilon^{\Phi}. \tag{11}$$

The cancellations are exact only when both observations enter the filter with the same modelling: same tropospheric mapping (which is why ΔT is small but not zero), same antenna PCV correction, and so on.

The receiver-bias cancellation of $B_{r,s}$ in Equation (10) holds for CDMA constellations, whose group delay is a single per-signal receiver term. GLONASS is the exception: its FDMA inter-frequency biases are per-FCN-slot states, so two satellites on different slots carry different receiver delays and the term does *not* cancel between them — a residual $B_{r,s}^{(k(i))} - B_{r,s}^{(k(p_g))}$ survives the single difference. Accordingly `BssdTransform` zeroes the receiver-DCB columns after differencing *except* the per-slot GLONASS IFB states (guarded by `StateKeys::isGlonassPerSlotIfbKey`), and `PivotManager` bars GLONASS from pivot election, for the same reason. GLONASS BSSD is default-OFF (`estimation.between_satellite_sd.enable_glonass`); when enabled, Equation (10) acquires the surviving inter-slot IFB difference as an additional estimated column.

Pivot selection. Election is not by elevation alone. The primary key is a *hard* lexicographic tier on phase-signal coverage: a full-coverage candidate outranks any partial-coverage candidate regardless of its score. Only within a coverage tier does the additive score $-\min\!\big(1, \sqrt{\sigma^2_{\hat{N},\max}}\big) + \mathrm{el}/90 - \frac{1}{2}\mathbb{1}[\mathrm{el} < 20^\circ]$ apply, in which float-ambiguity convergence $\sqrt{\sigma^2_{\hat{N},\max}}$ and elevation enter as co-equal terms (with a hard low-elevation guardrail below 20°). Layered on top is an ephemeris-freshness bar: a candidate whose broadcast $|\mathrm{obs} - \mathrm{toe}|$ exceeds its constellation's freshest qualified peer by more than `kPivotToeSpreadMaxS` (1200 s) is barred from *pivot* election (though still usable as a non-pivot), which prevents a stale-eph pivot from stepping every SD row it anchors. Candidates below `estimation.between_satellite_sd.pivot_min_elevation` are excluded outright. The baseline DD path scores differently: `DdPivotManager` ranks on $\mathrm{avg_snr} + 0.01 \cdot \mathrm{num_sigs}$ and does not use elevation.

Pivot transitions. A handover fires on two kinds of trigger. The visibility-driven one is simple — when the incumbent pivot leaves the visible set, p_g is re-elected from the remaining satellites — but the dominant one in practice is *event-driven*: the post-fit residual analyser calls `PivotManager::requestForcedHandover` when the pivot's *observation* slipped while its filter *state* is still intact (so the resting `isPivotDisturbed` check, which tests only that the pivot ambiguity sits at exactly $(0, 0)$, would not catch it). The forced request carries an exclusion set so the handover cannot bounce onto

another offset satellite, a `retired_pivots_this_epoch_` backstop that terminates cascades, a same-epoch re-entry gate (`hasPendingForcedHandover`), and a re-inflation set for states that absorbed a coherent pivot-side code bias (the information-conserving transform below cannot evict that offset on its own). The baseline DD path exposes the symmetric `DdPivotManager::requestForcedRerank`. This event response is load-bearing precisely because the pivot's states are frozen at zero variance (below): a stiff, time-linked zero-q quantity requires a wired jump response.

On election the float ambiguities $\{\Delta N_{r,s}^{i,p_g^{\mathrm{old}}}\}_{i,s}$ must be re-expressed relative to p_g^{new}. Where the new pivot already carries an established ambiguity in the group, the engine applies the linear S-transformation $\boldsymbol{x}^{\mathrm{new}} = \boldsymbol{S}\,\boldsymbol{x}^{\mathrm{old}}$ with $\boldsymbol{S}$ chosen so that $\Delta N_{r,s}^{p_g^{\mathrm{new}},p_g^{\mathrm{old}}}$ becomes the new reference (zero), preserving the integer information accumulated since the last fix; the covariance transforms as $\boldsymbol{P}^{\mathrm{new}} = \boldsymbol{S}\,\boldsymbol{P}^{\mathrm{old}}\,\boldsymbol{S}^{\mathrm{T}}$. When it cannot — the new pivot has no existing ambiguity in the $(\mathrm{cons}, \mathrm{code})$ group (`DdPivotManager`, logged as "reset-and-reseed"), or a disturbed pivot has no healthy replacement in a single-satellite constellation (`PivotManager::handleDisturbedPivot` re-zeroes $\boldsymbol{x}$ and the corresponding rows/columns of $\boldsymbol{P}$ in place) — the transform is skipped and the stale rows are discarded and re-seeded. In the uncombined PPP-RTK parameterisation the same $\boldsymbol{S}\,\boldsymbol{P}\,\boldsymbol{S}^{\mathrm{T}}$ re-reference is applied not only to the ambiguity block but to *three* state groups — `reReferenceAmbiguities`, `reReferenceIonosphere`, and `reReferenceSatBiases` — which is why `BssdTransform` zeroes the pivot's ionosphere and sat-bias $\boldsymbol{H}$-columns after differencing, and why $\Delta I_{r,1}^{i,p_g}$ in Equations (10) and (11) is itself a pivot-referenced SD quantity, not an absolute slant delay. The S-basis constraint is realised operationally by freezing the pivot's ambiguity, ionosphere, and sat-bias covariances to exactly zero (and suppressing their process noise) — the very condition `isPivotDisturbed` tests against.

`PivotManager::getCleanHandovers` classifies each transition as lossless-invertible (*clean*) or discontinuous (disturbed/forced/re-seeded), so the RTS smoother can inverse-reference across a clean handover rather than force a segment boundary; the losslessness claim above holds only for the clean subset. These are two distinct code paths sharing the algebra above: in PPP-RTK the pivot re-election and the "sticky-pivot" retention policy live in `filter/PivotManager`, whereas the baseline DD path performs the analogous re-reference through the ambiguity-rename S-transform in `filter/DdPivotManager`, governed by the `baseline.enable_pivot_handover` flag.

4.2 Between-receiver single difference (BRSD)

In baseline modes there is at least one rover r_{rov} and exactly one fixed-coordinate reference r_{ref}. The BRSD operator forms

$$\nabla P^{i}_{r_{\mathrm{rov}}r_{\mathrm{ref}},s} \;:=\; P^{i}_{r_{\mathrm{rov}},s} - P^{i}_{r_{\mathrm{ref}},s}, \tag{12}$$

and analogously for $\nabla\Phi$. The cancellations produced by $\nabla\cdot$ are dual to those produced by $\Delta\cdot$: BRSD cancels *satellite*-side common terms ($c\,\mathrm{d}t^i$, B_s^i, b_s^i) but preserves receiver clock and biases of *both* stations. No state-space estimation of satellite biases is required when ∇ is applied — they disappear before reaching the filter.

4.3 Double difference (DD)

The composition $\nabla\Delta\cdot = \Delta\big(\nabla\,\cdot\big)$ is order-independent up to noise correlation: $\Delta\big(\nabla\,\cdot\big) = \nabla\big(\Delta\,\cdot\big)$. Applying it to the carrier-phase model and collecting common-mode cancellations:

$$\nabla\Delta\Phi^{i,p_g}_{r_{\mathrm{rov}}r_{\mathrm{ref}},s} \;=\; \nabla\Delta\rho^{i,p_g}_{r_{\mathrm{rov}}r_{\mathrm{ref}}} + \nabla\Delta T^{i,p_g}_{r_{\mathrm{rov}}r_{\mathrm{ref}}} - \mu_s\,\nabla\Delta I^{i,p_g}_{r_{\mathrm{rov}}r_{\mathrm{ref}},1} + \lambda_s\,\nabla\Delta N^{i,p_g}_{r_{\mathrm{rov}}r_{\mathrm{ref}},s} + \nabla\Delta\varepsilon^{\Phi}. \tag{13}$$

What survives in Equation (13) is precisely the geometry term (linear in the rover position perturbation), the differential atmospheric delays, and the integer DD ambiguity — the structural form that makes carrier-phase RTK possible.

Short vs. long baselines. On short baselines ($\lesssim 10\,\mathrm{km}$) the differential atmospheric terms in Equation (13) are negligible at the millimetre level and need not be modelled. As baseline length grows, the differential ionosphere $\nabla\Delta I$ scales linearly with separation and saturates at typically 1 m for separations of a few hundred kilometres, swamping the carrier wavelength. Two mitigations are available in `GnssProcessor`:

- `baseline.use_iono_free` forms the dual-frequency ionosphere-free combination at the DD level, eliminating the first-order ionospheric term at the cost of a non-integer ambiguity (AR is automatically excluded for IF rows) and $\sim 3\times$ amplified code noise;
- `baseline.estimate_differential_tropo` adds a single $\Delta\mathrm{ZWD}_{r_{\mathrm{rov}} r_{\mathrm{ref}}}$ state with slow random-walk dynamics, capturing the residual differential wet delay that survives the BRSD on baselines with significant altitude difference.

4.4 S-transform structure

Both BSSD and DD admit a matrix-form representation as $\boldsymbol{x}_{\mathrm{diff}} = \boldsymbol{S}\,\boldsymbol{x}_{\mathrm{undiff}}$ with $\boldsymbol{S}$ acting as a block design transform. For BSSD restricted to the ambiguity block of one constellation g on one signal s, with $|\mathcal{S}_g|$ tracked sats, $\boldsymbol{S}$ is a $(|\mathcal{S}_g|-1)\times|\mathcal{S}_g|$ matrix of the form

$$\boldsymbol{S}_{\mathrm{BSSD}} \;=\; \begin{bmatrix} 1 & 0 & \cdots & 0 & -1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 & -1 & 0 & \cdots & 0 \\ \vdots & & \ddots & & \vdots & & & \\ 0 & 0 & \cdots & 0 & -1 & 0 & \cdots & 1 \end{bmatrix}, \tag{14}$$

where the column of -1's sits at the index of the pivot p_g. The DD operator is the *composition* of the two single-difference operators applied in sequence, $\boldsymbol{S}_{\mathrm{DD}} = \boldsymbol{S}_{\mathrm{BSSD}}\,\boldsymbol{S}_{\mathrm{BRSD}}$, realised in code as `BrsdTransform::apply` followed by `BssdTransform::differenceBrsd` (`include/measurement/BssdTransform.hpp`). Only on a fully populated satellite $\times$ receiver observation grid does this composition coincide with the Kronecker product $\mathbf{D}_{\mathrm{sat}} \otimes \mathbf{D}_{\mathrm{rx}}$ of the between-satellite and between-receiver differencing matrices acting on the vectorised observations — a useful mental model, but not what the non-rectangular real-data case computes. The transformed covariance is $\boldsymbol{P}_{\mathrm{diff}} = \boldsymbol{S}\,\boldsymbol{P}_{\mathrm{undiff}}\,\boldsymbol{S}^{\mathrm{T}}$, a generally dense matrix even when $\boldsymbol{P}_{\mathrm{undiff}}$ is diagonal (because each differenced row borrows from the same pivot column). This dense correlation structure is the algebraic reason DD-based filters cannot be naively implemented as independent per-pair scalar updates — the correctly correlated joint update is what unlocks integer ambiguity recovery via the LAMBDA decorrelation. The general theory and a thorough catalogue of admissible S-transforms is given in Teunissen [4] and in the rank-defect treatment of Jonge [1].

Implementation notes. Two facts about how `GnssProcessor` realises the transforms above are worth stating explicitly. First, the chosen S-system is never materialised as an explicit minimum-constraint $\boldsymbol{S}$-basis matrix; it is realised *implicitly* through the pivot-differencing operator (`measurement/BssdTransform`), a faithful, S-basis-invariant realisation of the uncombined parameterisation of Odijk [2]. The three regimes — undifferenced, BSSD, and DD — are dispatched polymorphically by a `DifferencingMode` strategy in `measurement/MeasurementBuilder`, not by

inline config branches. Second, integer recovery on a DD row requires that all four contributing observations share the same RINEX signal code. The row is marked `integer_capable` only when all three conjuncts hold: `!brsd_aliased` on *both* the i_i and pivot BRSD entries (rover and reference agree on each, i.e. no cross-attribute alias) *and* the rover uses the same code on i_i as on the pivot ($\texttt{rov_code}_i = \texttt{rov_code}_{\text{piv}}$). The third conjunct is independent of the `brsd_aliased` flags: a row can be un-aliased on both entries yet still carry a non-integer DCB/IFB residual if the rover mixes codes across i_i and the pivot. Rows failing any conjunct are carried as float and excluded from ambiguity resolution.

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